

Performance Metrics and Calibration for Bayesian Linear Regression

ELG 5218 - Uncertainty Evaluation in Engineering Measurements and Machine Learning

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One prediction: a temperature and its uncertainty

Suppose Bayesian linear regression predicts a future temperature at input $\mathbf{x}_{*i}$:

$$Y_{*i} \mid \mathbf{x}_{*i}, \mathcal{D} \sim \mathcal{N}(20, 2^2).$$

- ▶ Predictive mean $\hat{y}_{*i} = 20^\circ\text{C}$: the point prediction.
- ▶ Predictive standard deviation $s_{*i} = 2^\circ\text{C}$: the spread of possible observations.
- ▶ The actual temperature later observed is $y_{*i} = 22^\circ\text{C}$.

Connection to the regression lecture

$$\hat{y}_{*i} = \mathbf{x}_{*i}^\top \boldsymbol{\mu}_N, \quad s_{*i}^2 = \beta^{-1} + \mathbf{x}_{*i}^\top \boldsymbol{\Sigma}_N \mathbf{x}_{*i}.$$

Y_{*i} denotes the random future outcome; y_{*i} is its observed value. The predictive variance includes observation noise.

Throughout, N is the training-set size and N_{test} is the number of held-out test cases. The Gaussian formulas assume fixed or plug-in hyperparameters.

Which data are used for each operation?

Dataset	Role
$\mathcal{D}_{\text{train}}$	Fit the regression model, including its noise parameters. Its size is N . In the lecture, $\mathcal{D} = \mathcal{D}_{\text{train}}$.
Validation data / CV	Select features, hyperparameters, and any planned uncertainty-adjustment method when needed. This is part of model development.
$\mathcal{D}_{\text{cal}}$ (optional)	Fit the chosen post-hoc adjustment, such as a standard-deviation multiplier a . Its size is N_{cal} .
$\mathcal{D}_{\text{test}}$	Evaluate the frozen prediction procedure. Its size is N_{test} . No model or adjustment fitting on this set.

Without recalibration: fit on training data, then evaluate on test data.

With a calibration split: fit the model on training data, fit the adjustment on calibration data, then evaluate on test data.

These are data roles, not a requirement for four fixed subsets. Cross-validation can replace dedicated development splits. Respect temporal and group dependence.

The test set evaluates the final prediction procedure

- ▶ All averages over N_{test} in this deck describe **final test-set evaluation**: RMSE, NLPD, CRPS, PICP, mean width, PIT, and standardized residuals.
- ▶ Without recalibration, evaluate the predictive distribution produced directly by Bayesian linear regression.
- ▶ With scale recalibration, first estimate a from $\mathcal{D}_{\text{cal}}$ and freeze it. On test inputs, use

$$\hat{y}_{*i}^{\text{final}} = \hat{y}_{*i}, \quad s_{*i}^{\text{final}} = a s_{*i}.$$

Replace s_{*i} by s_{*i}^{final} in the Gaussian scores, CDFs, intervals, and diagnostics. RMSE is unchanged because the means are unchanged.

- ▶ Development diagnostics can use validation or calibration data, but they are not independent final performance estimates.

Do not use test results to choose an adjustment

If test results guide a model revision or recalibration, those data have become development data. Use a fresh test set or an appropriate nested evaluation procedure for the final assessment.

What do we want to evaluate?

Point accuracy Was the predicted temperature close to the observed temperature?

Calibration Across many predictions, did events assigned probability 90% occur approximately 90% of the time?

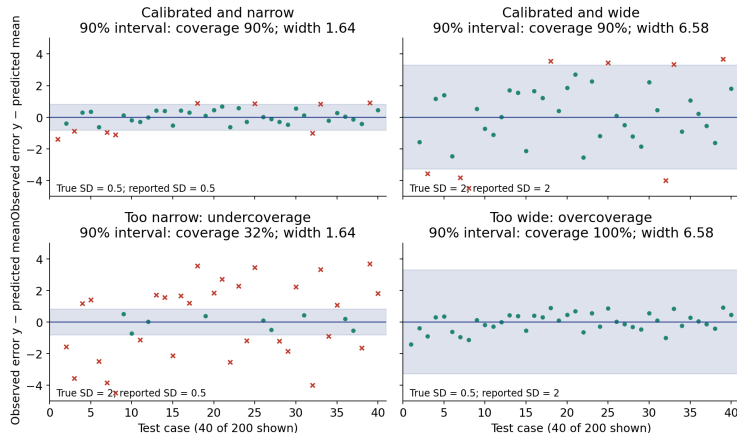
Sharpness How narrow were the predictive distributions or intervals?

Why accuracy alone is insufficient

Models reporting $\mathcal{N}(20, 0.5^2)$ and $\mathcal{N}(20, 5^2)$ both predict a mean of 20. When 22 is observed, both have error 2, but they made very different uncertainty claims.

RMSE evaluates point accuracy. NLPD and CRPS evaluate the complete predictive distribution. Coverage and PIT help diagnose calibration; interval width describes sharpness.

Pictures of calibration and sharpness



Each panel simulates 200 test cases; 40 are shown. For each case we plot its error $y_{*i} - \hat{y}_{*i}$, so every central 90% prediction interval $\hat{y}_{*i} \pm 1.645 s_{*i}$ becomes the band $\pm 1.645 s_{*i}$ around zero (here s_{*i} is the same for all cases in a panel). A green dot is an observation inside its interval; a red cross is a *miss*, an observation outside its interval. Coverage is the fraction of the 200 cases that are inside. The horizontal axis is only the case number.

What can make reported uncertainty too narrow or too wide?

The bottom-row mismatches in the previous figure are *simulated* (reported SD 0.5 vs true 2, and the reverse). In practice they arise from:

Too narrow (overconfident, undercoverage)

- ▶ Noise variance β^{-1} underestimated: β fixed too large, or estimated from training residuals of an overfit model (many features, small N).
- ▶ Model misspecification (missing nonlinearity or feature): Σ_N measures uncertainty *within* the model, not the error of using the wrong model.
- ▶ Test conditions noisier than training: new sensor, another season, extrapolation beyond the training inputs.
- ▶ Dependent data treated as independent: Σ_N too small. Heavy tails under a Gaussian likelihood.

Too wide (underconfident, overcoverage)

- ▶ Noise variance overestimated: β fixed too small, or $\hat{\beta}^{-1}$ inflated by outliers or an unusually noisy training period.
- ▶ Test conditions calmer than training: better sensor, calmer season.
- ▶ Model misfit absorbed into the noise estimate, applied to inputs where the model is actually accurate.

Both at once: with input-dependent noise, one constant β^{-1} is too narrow in noisy regions and too wide in quiet ones, even when aggregate coverage looks correct.

What does a proper score mean?

A score is a loss assigned to a prediction after the outcome becomes known.

The purpose of propriety

If the true outcome distribution is P , reporting P should give the lowest expected loss. Deliberately reporting a different distribution Q should not improve the expected score:

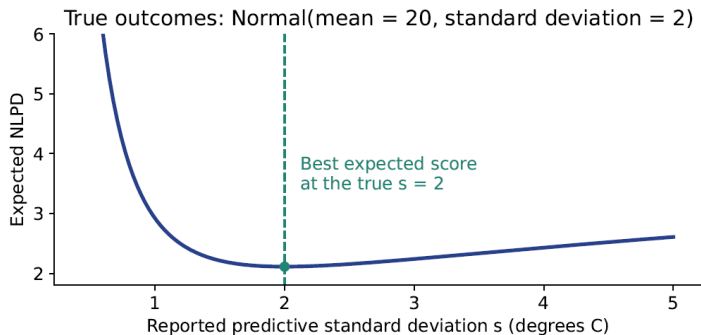
$$\mathbb{E}_{Y \sim P}[S(P, Y)] \leq \mathbb{E}_{Y \sim P}[S(Q, Y)].$$

A *strictly* proper score makes P the unique optimum.

- ▶ Example: if the true uncertainty is 2°C , the score should not reward systematically claiming 0.2 or 20 instead.
- ▶ This is a guarantee **in expectation over outcomes**, not for every individual observation.
- ▶ The true distribution is unknown in practice. Propriety is a property of the scoring rule, not proof that a fitted model is correct.

NLPD and CRPS are strictly proper distributional scores under standard conditions. Gneiting & Raftery (2007).

A proper score rewards the correct uncertainty



If $Y \sim \mathcal{N}(20, 2^2)$ and we report $\mathcal{N}(20, s^2)$, then

$$\mathbb{E}[\text{NLPD}] = \frac{1}{2} \log(2\pi s^2) + \frac{\mathbb{E}[(Y - 20)^2]}{2s^2} = \frac{1}{2} \log(2\pi s^2) + \frac{4}{2s^2}.$$

The minimum occurs at $s = 2$. Reporting either too little or too much uncertainty worsens the expected score.

Negative Log Predictive Density (NLPD)

For one observed test case, set $e_i = y_{*i} - \hat{y}_{*i}$. Then

$$\text{NLPD}_i = -\log p(y_{*i} \mid \mathbf{x}_{*i}, \mathcal{D}) = \underbrace{\frac{1}{2} \log(2\pi s_{*i}^2)}_{\text{penalty for large spread}} + \underbrace{\frac{e_i^2}{2s_{*i}^2}}_{\text{error relative to spread}}.$$
$$\overline{\text{NLPD}} = \underbrace{\frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{1}{2} \log(2\pi s_{*i}^2)}_{\text{spread part}} + \underbrace{\frac{R}{2}}_{\text{error part}}, \quad R = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{(y_{*i} - \hat{y}_{*i})^2}{s_{*i}^2}.$$

- ▶ **Small spread, large error:** dividing e_i^2 by a small s_{*i}^2 makes the second term large.
- ▶ **Excessively wide prediction:** the first term increases with s_{*i} . As $s_{*i} \rightarrow \infty$, the score also tends to infinity.
- ▶ **Very low density at the observation:** $-\log p$ becomes large as $p \rightarrow 0$. This is the same loss viewed directly through the predicted density.

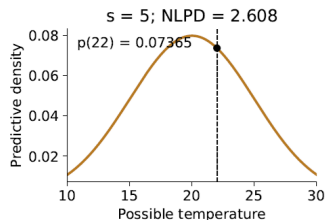
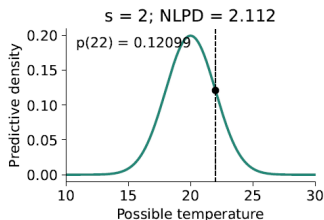
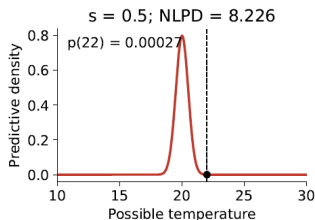
If $R \approx 1$ (errors are as large as claimed): $\overline{\text{NLPD}} \approx \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{1}{2} \log(2\pi e s_{*i}^2).$

NLPD looks at density where the observation occurs

Error $e_i = 2$, with all means equal to 20

s_{*i}	Log-spread term	Error term	Total NLPD _i
0.5	0.226	8.000	8.226
2	1.612	0.500	2.112
5	2.528	0.080	2.608

Same mean = 20; observed temperature = 22



The black vertical line marks the observed temperature 22. The black dot is the predictive density evaluated there.

- ▶ The narrow forecast assigns almost no density to 22 and receives a large loss.
- ▶ The very wide forecast spreads its density over many temperatures, reducing the density at 22.

NLPD can be negative because a density can exceed 1.

Continuous Ranked Probability Score (CRPS)

For this test input, the predictive CDF is

$$F_i(t) = P(Y_{*i} \leq t \mid \mathbf{x}_{*i}, \mathcal{D}) = \Phi\left(\frac{t - 20}{2}\right).$$

It answers: **what probability did we assign to a temperature at or below t ?**

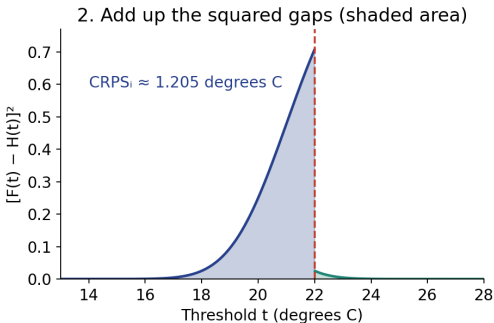
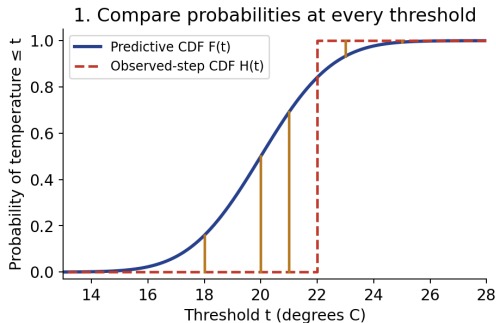
Threshold t	Predicted probability $F_i(t)$	Observed answer $\mathbf{1}\{22 \leq t\}$
18	0.159	0 (no)
20	0.500	0 (no)
22	0.841	1 (yes)
24	0.977	1 (yes)

Once 22 is observed, the answers form a step function:

$$H_i(t) = \mathbf{1}\{y_{*i} \leq t\} = \begin{cases} 0, & t < 22, \\ 1, & t \geq 22. \end{cases}$$

F_i is the model's CDF for this input, not the empirical CDF of the training targets.

One CRPS value is an area under the squared CDF gap



$$CRPS_i = \int_{-\infty}^{\infty} [F_i(t) - H_i(t)]^2 dt = \int_{-\infty}^{y_{*i}} F_i(t)^2 dt + \int_{y_{*i}}^{\infty} [1 - F_i(t)]^2 dt.$$

The orange segments on the left are probability errors. Square these gaps and integrate over thresholds to obtain the shaded area on the right. Here it is about 1.205 °C.

What the CRPS area means: one yes/no question per threshold

Each threshold t asks: “Will the temperature be $\leq t$?” $F_i(t)$ is the forecast’s probability of “yes”; $H_i(t)$ is the true answer after observing $y_{*i} = 22$; $[F_i(t) - H_i(t)]^2$ is the penalty: 0 if sure and right, 1 if sure and wrong.

Threshold t	16	18	20	21	just below 22	23	24
Forecast $F_i(t)$	0.02	0.16	0.50	0.69	0.83	0.93	0.98
Answer $H_i(t)$	0	0	0	0	0	1	1
Penalty	0.00	0.03	0.25	0.48	0.69	0.00	0.00

For $\mathcal{N}(20, 2^2)$, CRPS adds the penalties along the temperature axis: $\text{CRPS}_i = \int [F_i(t) - H_i(t)]^2 dt = 1.205^\circ\text{C}$.

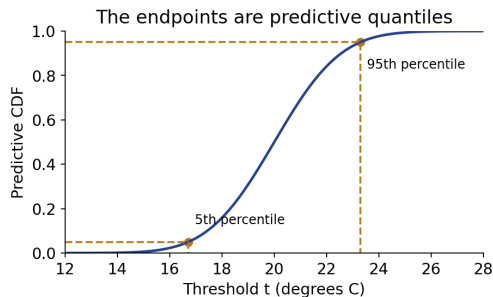
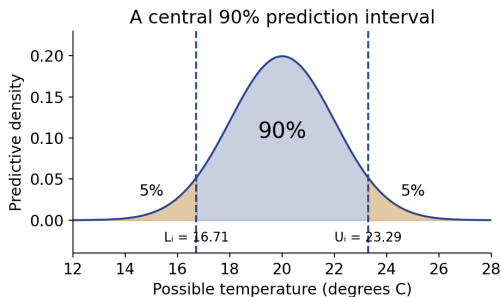
Where the area comes from

- ▶ **Far away** (below 16, above 24): confidently right, no penalty.
- ▶ **Between 20 and 22**: the forecast leaned “yes”, the answer was “no”. Most of the area.
- ▶ **Near the centre**: $F \approx 0.5$ costs ≈ 0.25 per degree wherever y lands: the price of spread.

A distance between forecast and outcome

- ▶ Observed 22: $\mathcal{N}(20, 0.5^2) \rightarrow 1.72$, $\mathcal{N}(20, 2^2) \rightarrow 1.21$, $\mathcal{N}(20, 5^2) \rightarrow 1.48$. Too sure or too vague both cost.
- ▶ In $^\circ\text{C}$, **lower is better**; average over test cases like a mean absolute error.

What are the endpoints of a central prediction interval?



Central 90% means leaving equal probability in the two tails: 5% below and 5% above.

$$L_i(c) = F_i^{-1}\left(\frac{1-c}{2}\right), \quad U_i(c) = F_i^{-1}\left(\frac{1+c}{2}\right).$$

L_i and U_i are predicted **lower and upper temperatures**, in the same units as the observation. They are probabilistic bounds, not physical limits on what can occur.

From predictive standard deviation to interval endpoints

For a Gaussian prediction, let $a_c = \Phi^{-1}((1 + c)/2)$:

$$L_i(c) = \hat{y}_{*i} - a_c s_{*i}, \quad U_i(c) = \hat{y}_{*i} + a_c s_{*i}.$$

Our prediction: mean 20, standard deviation 2

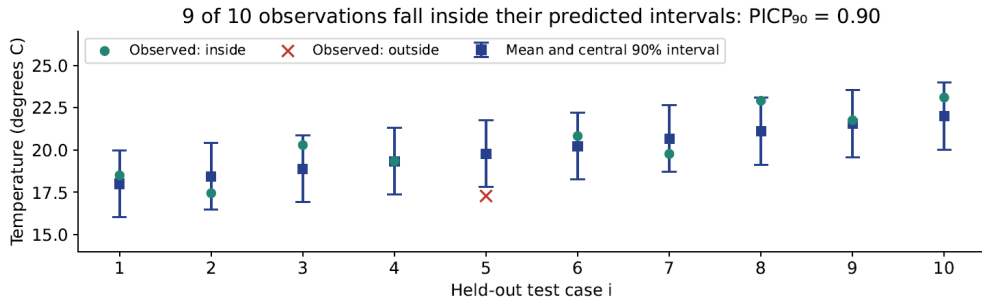
$$c = 0.90 : \quad a_c \approx 1.645, \quad [L_i, U_i] = [16.71, 23.29],$$

$$c = 0.95 : \quad a_c \approx 1.960, \quad [L_i, U_i] = [16.08, 23.92].$$

The observation 22 lies inside both intervals. An observation of 24 would lie outside both.

- ▶ Requesting higher probability produces a wider interval.
- ▶ A 90% interval permits misses: even a correct model misses about 10% of observations in repeated use.
- ▶ These intervals concern a future *observation*. A credible interval for its latent mean excludes the observation-noise term and is generally narrower.

Prediction Interval Coverage Probability (PICP)



$$\text{PICP}(c) = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \mathbf{1}\{L_i(c) \leq y_{*i} \leq U_i(c)\}.$$

Each indicator is 1 for a covered observation and 0 for a miss. Averaging converts the count into a fraction, which we compare with the claimed probability c . This tiny example illustrates the calculation. Ten cases are insufficient to establish calibration; observed coverage varies between samples.

What do coverage and width tell us?

Coverage asks whether the claimed frequency is credible

For nominal 90% intervals: 60% coverage means too many misses; 100% coverage on a large representative sample may indicate unnecessarily conservative intervals. Bias and incorrect tails can also affect coverage.

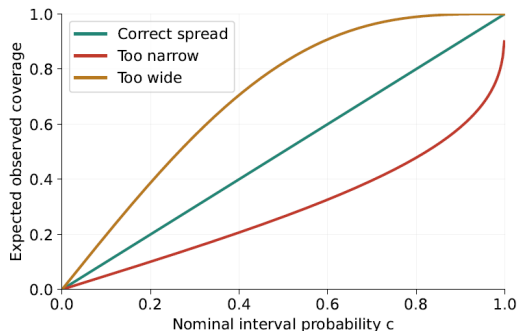
Mean Prediction Interval Width (MPIW) asks how precise the forecasts are

$$\text{MPIW}(c) = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} [U_i(c) - L_i(c)].$$

Our central 90% temperature interval has width $23.29 - 16.71 = 6.58$ °C.

- ▶ Very wide intervals can cover almost everything while giving little useful precision.
- ▶ Very narrow intervals can look precise while missing most outcomes.
- ▶ Seek narrow intervals *with reliable coverage*. Check multiple probability levels and input regions, not just one overall coverage fraction.

A reliability curve checks several interval probabilities



For each c , build intervals and compute $\text{PICP}(c)$.

- ▶ Horizontal axis: probability the model claims.
- ▶ Vertical axis: fraction actually covered.
- ▶ Diagonal: observed frequency matches the claim.
- ▶ Below diagonal: undercoverage.
- ▶ Above diagonal: overcoverage.

Example: at $c = 0.9$, a value of 0.6 means only 60% of nominal 90% intervals cover their observations.

Curves show idealized Gaussian scale errors. A finite test set gives a noisy curve. Aggregate agreement can hide errors in particular groups or input regions.

PIT: where did the observation land in its predicted distribution?

$$u_i = F_i(y_{*i}) = \Phi\left(\frac{y_{*i} - \hat{y}_{*i}}{s_{*i}}\right).$$

For our forecast $\mathcal{N}(20, 2^2)$ and observation 22,

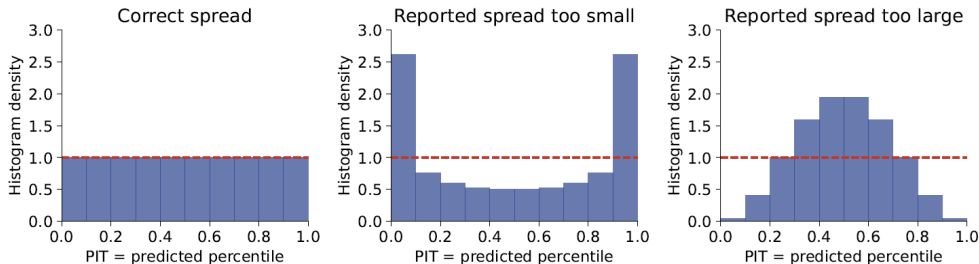
$$u_i = \Phi(1) \approx 0.841.$$

The observed temperature lies at the **84th predicted percentile**: the model assigned about 84% probability to temperatures at or below it.

- ▶ For final assessment, repeat this for every test observation using its final predictive CDF.
- ▶ Under correct continuous predictive distributions, observations populate all predicted percentiles uniformly.
- ▶ About 10% should lie in each percentile band: 0–10%, 10–20%, and so on.

Here u_i denotes a PIT value, not the eigenvector coordinate used in the earlier Gaussian-geometry example.

Probability Integral Transform (PIT)



- ▶ **Correct spread:** outcomes fill the predicted percentiles approximately evenly.
- ▶ **Too narrow:** many outcomes lie in the forecast's extreme tails; PIT values collect near 0 and 1.
- ▶ **Too wide:** outcomes stay near the forecast's center; PIT values collect near 0.5.

These examples isolate Gaussian scale errors with correct means. Bias, wrong tail shape, or mixtures can complicate the patterns. A flat histogram alone does not prove conditional calibration or independence.

Standardized residuals: error measured in predicted standard deviations

$$\tilde{r}_{*i} = \frac{y_{*i} - \hat{y}_{*i}}{s_{*i}}, \quad R = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \tilde{r}_{*i}^2.$$

For error 2, a forecast standard deviation of 0.5 gives $\tilde{r}_{*i} = 4$; a standard deviation of 2 gives $\tilde{r}_{*i} = 1$; a standard deviation of 5 gives $\tilde{r}_{*i} = 0.4$.

- ▶ Correct Gaussian predictive marginals give standardized errors with mean 0 and second moment 1.
- ▶ R averages squared errors expressed in units of predicted uncertainty. A value well above 1 can flag underestimated uncertainty or bias.
- ▶ R is not generally the variance: it also includes squared mean bias.
- ▶ $R \approx 1$ is only one check. It does not establish the correct distribution shape.

Training residuals are not a fair predictive calibration test

Use test predictions for final assessment; use out-of-fold predictions for development. Fitting and checking on the same responses can make errors look artificially small.

Optional recalibration: fit the scale on calibration data

Use this step only when fitting an additional uncertainty adjustment. Fit the base model on $\mathcal{D}_{\text{train}}$ and freeze it. For each calibration case j :

$$\hat{y}_{\text{cal},j} = \mathbf{x}_{\text{cal},j}^{\top} \boldsymbol{\mu}_N, \quad s_{\text{cal},j}^2 = \beta^{-1} + \mathbf{x}_{\text{cal},j}^{\top} \boldsymbol{\Sigma}_N \mathbf{x}_{\text{cal},j}.$$

Fit one positive multiplier, holding predictive means fixed

Minimizing Gaussian NLPD on $\mathcal{D}_{\text{cal}}$ gives

$$\hat{a}^2 = \frac{1}{N_{\text{cal}}} \sum_{j=1}^{N_{\text{cal}}} \frac{(y_{\text{cal},j} - \hat{y}_{\text{cal},j})^2}{s_{\text{cal},j}^2}.$$

Then freeze $\hat{a}$. For test case i , use $s_{*i}^{\text{final}} = \hat{a} s_{*i}$ in scores and intervals.

- ▶ Calibration-set fit describes the adjustment; test-set performance assesses whether it generalizes.
- ▶ Scaling can correct global dispersion, but cannot repair a biased mean or incorrect distribution shape.

Changing β also changes the weight posterior, so it differs from scaling a frozen model's predictive standard deviation. A fold-based recalibration scheme can avoid a dedicated calibration split.

Summary of the questions answered by each measure

Kind	Measure	Question
Point accuracy	RMSE	How large are the point-prediction errors?
Proper distributional score	NLPD	How much density did the forecast assign to the outcome that occurred?
	CRPS	How close were the predicted threshold probabilities to the observed yes/no answers?
Calibration diagnostic	PICP	What fraction of observations fell inside their prediction intervals?
	Reliability	Does coverage match the claim at every interval probability c ?
	PIT	Did outcomes occupy predicted percentiles as frequently as expected?
	R	How large were squared errors relative to predicted variances?
Sharpness	MPIW	How wide were the intervals, in the units of the outcome?

On the untouched test set: **rank** models with a proper distributional score (NLPD and/or CRPS), report RMSE for the means, and use calibration diagnostics with width to **explain** the ranking. Recalibration, if used, must already be fitted.

References for the dataset distinction

Standard Bayesian linear regression

C. E. Rasmussen and C. K. I. Williams (2006), *Gaussian Processes for Machine Learning*, Sec. 2.1, especially Eqs. (2.7)–(2.9). The posterior and predictive distribution follow from the training data and model assumptions; there is no mandatory post-hoc calibration stage.

<https://gaussianprocess.org/gpml/chapters/RW2.pdf>

Optional recalibration of an existing predictor

V. Kuleshov, N. Fenner, and S. Ermon (2018), *Accurate Uncertainties for Deep Learning Using Calibrated Regression*:

- ▶ Sec. 3.2 (printed p. 2800): recommends a separate calibration set to reduce overfitting and describes a K -fold alternative.
- ▶ Sec. 3.5 (printed p. 2800): recommends diagnostic data distinct from training and calibration data.
- ▶ Table 1: experiments include Bayesian linear regression.

<https://proceedings.mlr.press/v80/kuleshov18a.html>

References and interpretation of the figures

- ▶ T. Gneiting and A. E. Raftery (2007). Strictly proper scoring rules, prediction, and estimation. *JASA*, 102(477), 359–378.
<https://doi.org/10.1198/016214506000001437>
- ▶ T. Gneiting, F. Balabdaoui, and A. E. Raftery (2007). Probabilistic forecasts, calibration and sharpness. *JRSS B*, 69(2), 243–268.
<https://doi.org/10.1111/j.1467-9868.2007.00587.x>
- ▶ V. Kuleshov, N. Fenner, and S. Ermon (2018). Accurate uncertainties for deep learning using calibrated regression. *ICML*, PMLR 80, 2796–2804. Sec. 3.2: calibration split or folds; Sec. 3.5: independent diagnostics. Table 1 includes Bayesian linear regression.
<https://proceedings.mlr.press/v80/kuleshov18a.html>

All figures are mathematical illustrations or simulated examples, not experimental results. The four calibration/sharpness panels use stratified normal quantiles to make their coverage comparison clear. Real test-set coverage fluctuates.

The same mathematical scores apply beyond Gaussian predictions, using the actual predictive density, CDF, and quantiles. The Gaussian closed forms are specific to this lecture's assumptions.

Acknowledgements/ Document preparation

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